

August 21, 2026

Dear Canadian Wildlife Federation Hiring Team,

I am writing to apply for the Junior Geospatial Developer position with the Canadian Wildlife Federation. With a Master of Science in Computer Science from the University of Calgary and several years of hands-on experience building geospatial software, spatial data workflows, and Earth observation data systems, I am excited by the opportunity to support freshwater conservation and ecosystem restoration through practical geospatial technology.

My graduate research focused on scalable management and analysis of large geospatial and satellite-imagery datasets. As a Research Software Developer with BigGeo/Mitacs at the University of Calgary, I built C++ and Python software for processing, encoding, storing, and retrieving Earth observation data. I automated SAR and optical imagery preprocessing using GDAL, Rasterio, SNAP, and satellite data APIs, and developed DGGS-based workflows that transformed imagery into structured tensors for multiresolution retrieval and analysis. This work is closely aligned with the role's focus on spatial data processing, geospatial modelling frameworks, and reliable data workflows.

I also developed a deep-learning framework for digital elevation model super-resolution, where I used Python, PyTorch, Rasterio, Google Earth Engine, and Streamlit to collect, preprocess, evaluate, and visualize terrain data. The project required careful documentation, data-quality checks, terrain-aware metrics, and clear communication of modelling assumptions and limitations. I understand how important it is for geospatial tools to be technically sound, reproducible, and understandable to the scientists and domain experts who depend on them.

Beyond research, I bring backend development experience with Django REST APIs, PostgreSQL, authentication workflows, and service integration, along with Git, Linux, Docker, AWS-based coursework, and GitHub Actions. While I would continue building depth in tools such as PostGIS, ArcGIS Enterprise, and web-mapping libraries, my background in Python, SQL, APIs, spatial data, documentation, and applied geospatial research gives me a strong foundation for contributing quickly and learning the team's stack carefully.

What draws me to CWF is the chance to apply software engineering to conservation work with real ecological impact. I would be grateful for the opportunity to help build and maintain geospatial tools that make complex environmental data more useful for decision-making, restoration planning, and collaboration across technical and scientific teams.

Thank you for considering my application. I look forward to the opportunity to discuss how my geospatial software, data-processing, research, and documentation experience can contribute to the Canadian Wildlife Federation.

Sincerely,
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